Coordinate systems and Projections

By

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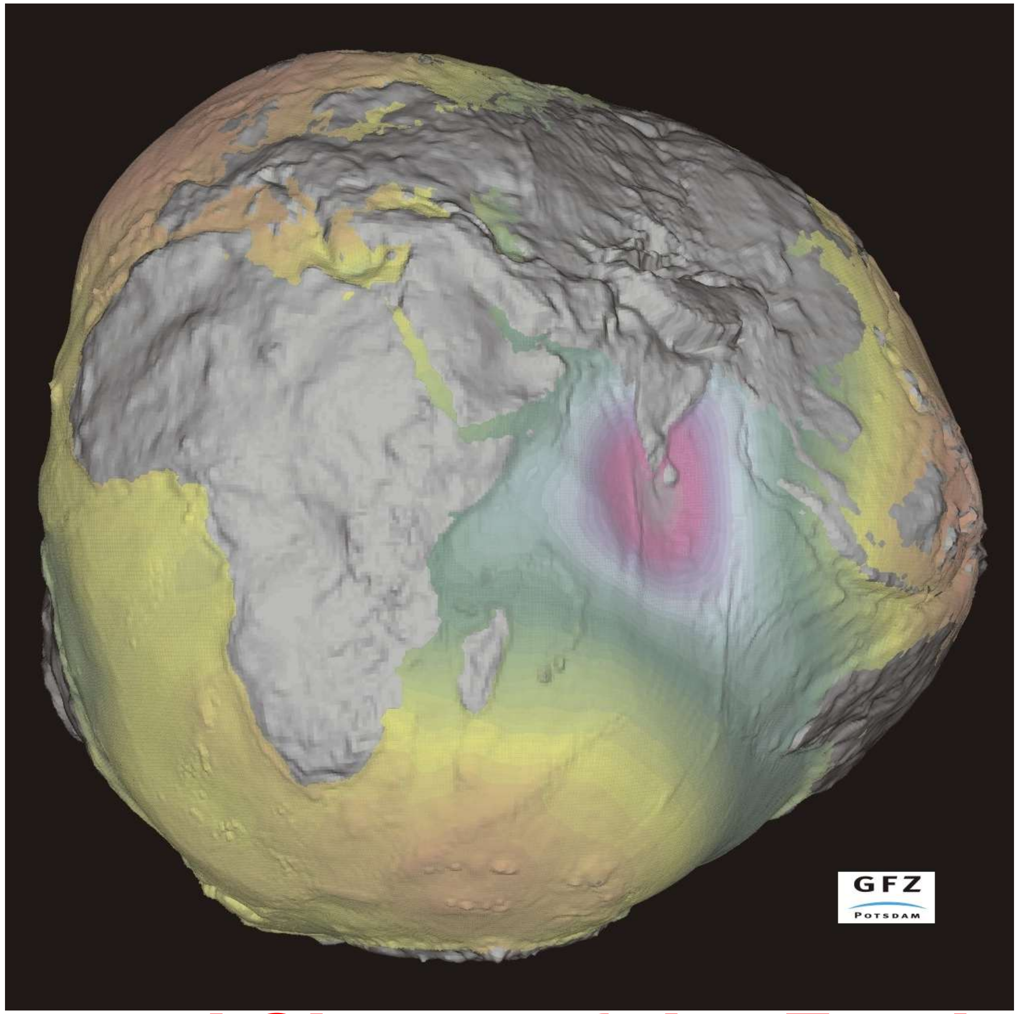
EARTH AS WE SEE FROM SPACE



Actual Shape of the Earth

Mapping: Local perspective vs Global perspective

- For smaller area, the task of mapping is relatively simple. The distances can be measured and plotted on the paper.

- **But, while mapping the large area or the entire world, the problem starts.**

- The **curvature of the Earth** complicates the task of mapping.
- Knowledge of Shape and Size of the Earth is indispensable if we are to make maps of its surface.

- This is what the geodesy deals with.

Geodesy

Geodesy is the science of measuring and monitoring the size and shape of the Earth and the location of points on its surface.

Survey

- **Plane Surveying (small area):** For small area, the surface of the Earth is assumed as a plane surface ignoring the curvature of the Earth. This is termed as Plane Surveying.
- **Geodetic Surveying (large area):** In these surveys, the curvature of the Earth is taken into account and a higher degree of accuracy in linear as well as angular observation is achieved.

What do we need to prepare the map ?

Requirements

- For surveying and mapping, most of the measurements are taken on the Earth surface.
- The surface of the Earth is an irregular surface. Directly using these measured values for further computations of distances, area, volume etc. will give wrong results, because the irregular surface of the Earth is not mathematical.
- We need a mathematical surface which can closely represent the Earth. All the measurements can then be reduced to that surface for computations.

We need a Model of the Earth.

Requirements

- Once the Model of the Earth is decided, we need to locate the positions of the features.
- **Everything is Relative**
- Example: Height of a point (**Height above What ?**)
- Distance of a place (**From Where ?**)
- **We need a Coordinate System / Reference System / Datum for defining the positions**

Requirements

- The positions of features on the Earth can be defined on a particular Reference System using appropriate Reference Ellipsoid.
- But these positions are 3 Dimensional
  - (Latitude, Longitude and Height) or (X, Y, Z)
- **We need a method to represent the 3D curved surface on a 2D surface (Plane paper, computer screen etc)**
- **That method is Map Projection**

Shape of the earth

- Throughout history, the shape of the Earth has been debated by scientists and philosophers.
- By 500 B.C. most scholars thought the Earth was completely spherical.
- The Greek philosopher Aristotle (384-322 B.C.) is credited as the first person to try and calculate the size of the Earth by determining its circumference (the length around the equator) He estimated this distance to be about 45,500 miles.
- Around 250 B.C., another Greek philosopher, Eratosthenes, computed the circumference of the Earth to be about 25,000 miles

- Today we know that the Earth is not perfectly spherical.

- The radius of curvature at poles is less than that at the Equator.

If the Earth has to be represented by a mathematical figure, it can be best represented by an Ellipsoid

Reference Ellipsoid

- An **Ellipsoid** is the mathematical surface which can be used to represent the Earth.

- The Reference Ellipsoid has its Centre a close approximation of center of mass of Earth and minor axis is parallel to axis of rotation of Earth

- Latitude, Longitude and Ellipsoidal Heights are defined on Reference Ellipsoid.

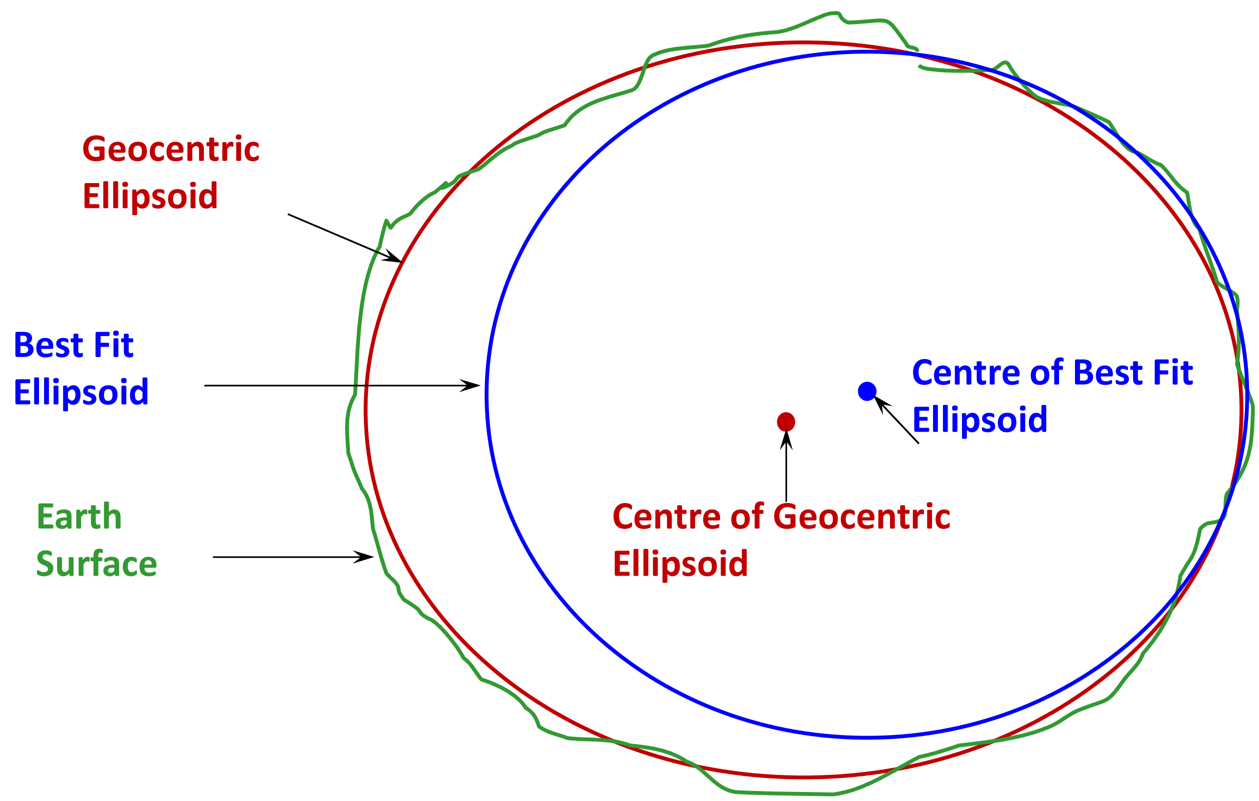
Reference Ellipsoids are of two types:

- **Best Fit Ellipsoid**

  This ellipsoid is based on the measurements within a region so it best fits that region only. The centre of such reference ellipsoid does not coincide with the centre of gravity of the Earth. An example of such type of ellipsoid is **Everest Ellipsoid**, which is used in India.

- **Geocentric Ellipsoid**

  As the name suggests, the centre of geocentric ellipsoid coincides with the centre of the Earth. This type of the ellipsoid can be used worldwide. An example of this type of ellipsoid is **WGS 84 Ellipsoid**.

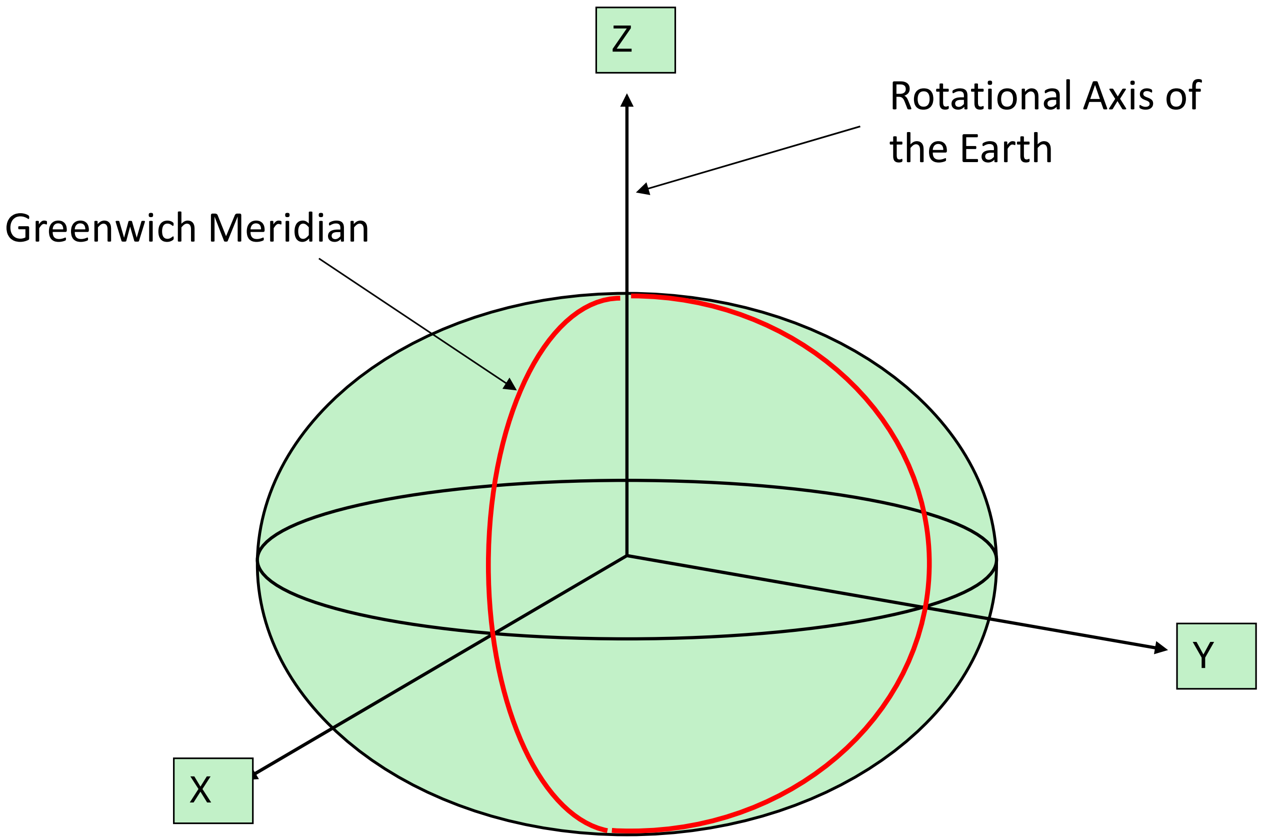


Best Fit Ellipsoid and Geocentric Ellipsoid

COORDINATE SYSTEM / REFERENCE SYSTEM

- Definition of the position of any point requires coordinates of that point on a well defined Coordinate System or Reference System

- In case of mapping, the Reference System needs to have a correlation with the Earth



Cartesian Co-ordinate System

Datum

A datum is a reference from which the measurements are made.

It can be thought of as a foundation on which all the structure of the map stands.

In the process of surveying/mapping two types of measurements are taken

- Horizontal
- Vertical

- Although the ellipsoid can be used for horizontal as well as vertical measurements, for practical reasons Geoid/MSL are generally used for vertical measurements.

DATUM USED IN SURVEYING

- **Horizontal Datum- (Ellipsoid)**
  - Horizontal datum is used for describing a point on the earth’s surface, in latitude and longitude or another coordinate system.
- **Vertical datum – (Geoid/MSL)**
  - Vertical datum is used to measure elevations or depths.

Classification of Reference Systems/DATUMS

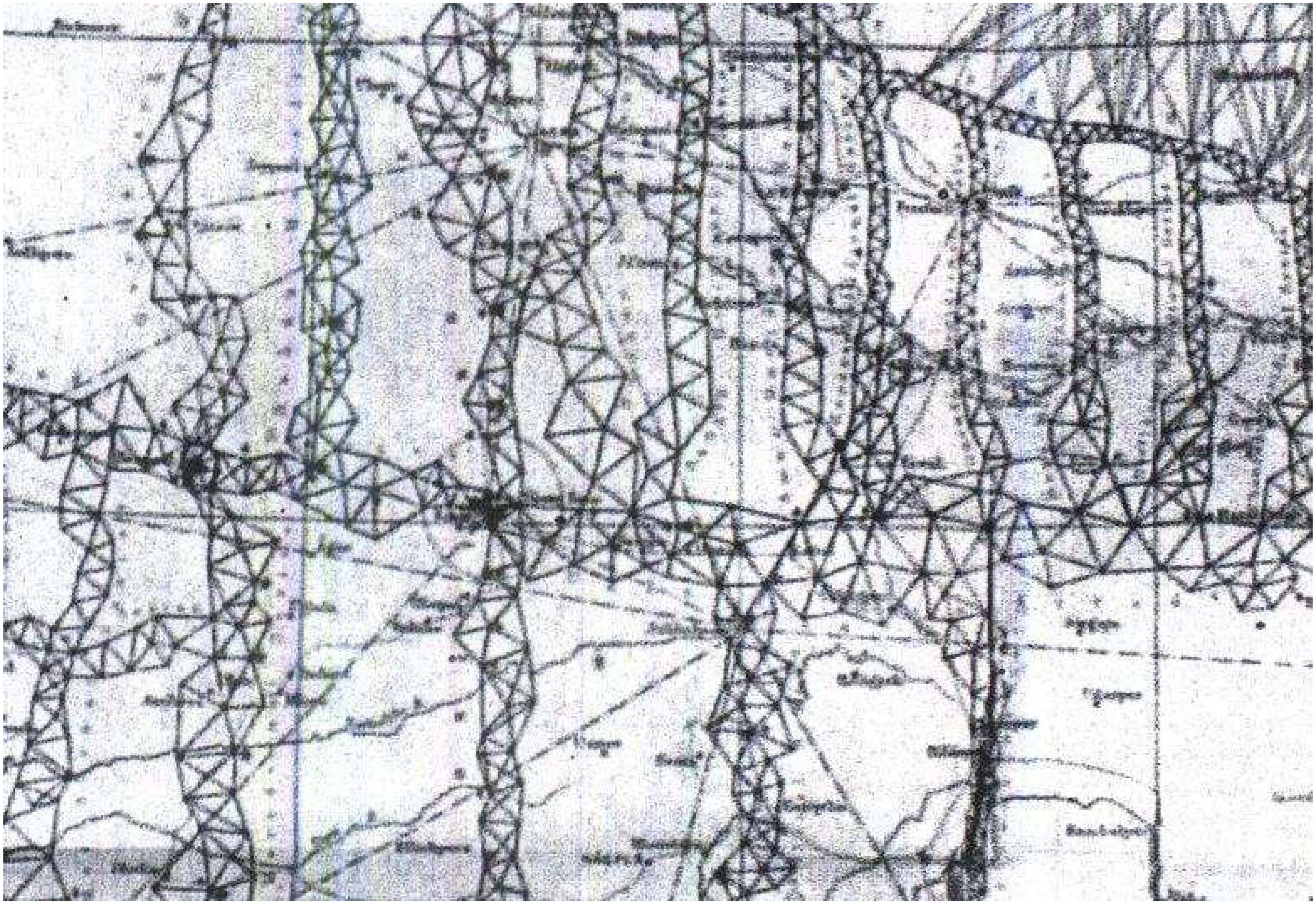
1. Local Geodetic Reference System
   - Indian Datum (Everest)

2. Global Geodetic Reference System
   - GRS 80
   - WGS84
   - ITRS and ITRF

1. Local Geodetic Reference System

- A local geodetic reference system is defined as a system for determining the latitude and longitude, based on an origin point whose latitude and longitude are determined by astronomical observation and the ellipsoid adopted by that nation.
- The datums realized by various countries prior to the advent of space geodetic techniques were local datums.
- They had the reference ellipsoids which were locally best fitting. The reason was that these were based on the observations taken in that region only.
- Everest system used in India till now is such type of reference system.

CALCUTTA
BOMBAY
MADRAS
BAY OF BEN
S E A
INDEX CHART
TO THE
GREAT TRIGONOMETRICAL SURVEY
OF
INDIA
SHOWING COLONEL LAMBTON'S NET WORK OF TRIANGULATION IN SOUTHERN INDIA
THE MERIDIONAL AND LONGITUDINAL CHAINS OF PRINCIPAL TRIANGLES.
THE BASE LINES MEASURED WITH THE COLBY APPARATUS.
THE LINES OF THE SPIRIT LEVELLING OPERATIONS.
THE ASTRONOMICAL, PENDULUM & TIDAL STATIONS.
THE LONGITUDINAL ARCS.
AND THE SECONDARY TRIANGULATION TO FIX THE PEAKS OF
THE HIMALAYAN & THE SULIMAN RANGES.
AND THE POSITIONS OF BANGKOK, KANDAHAR &c.



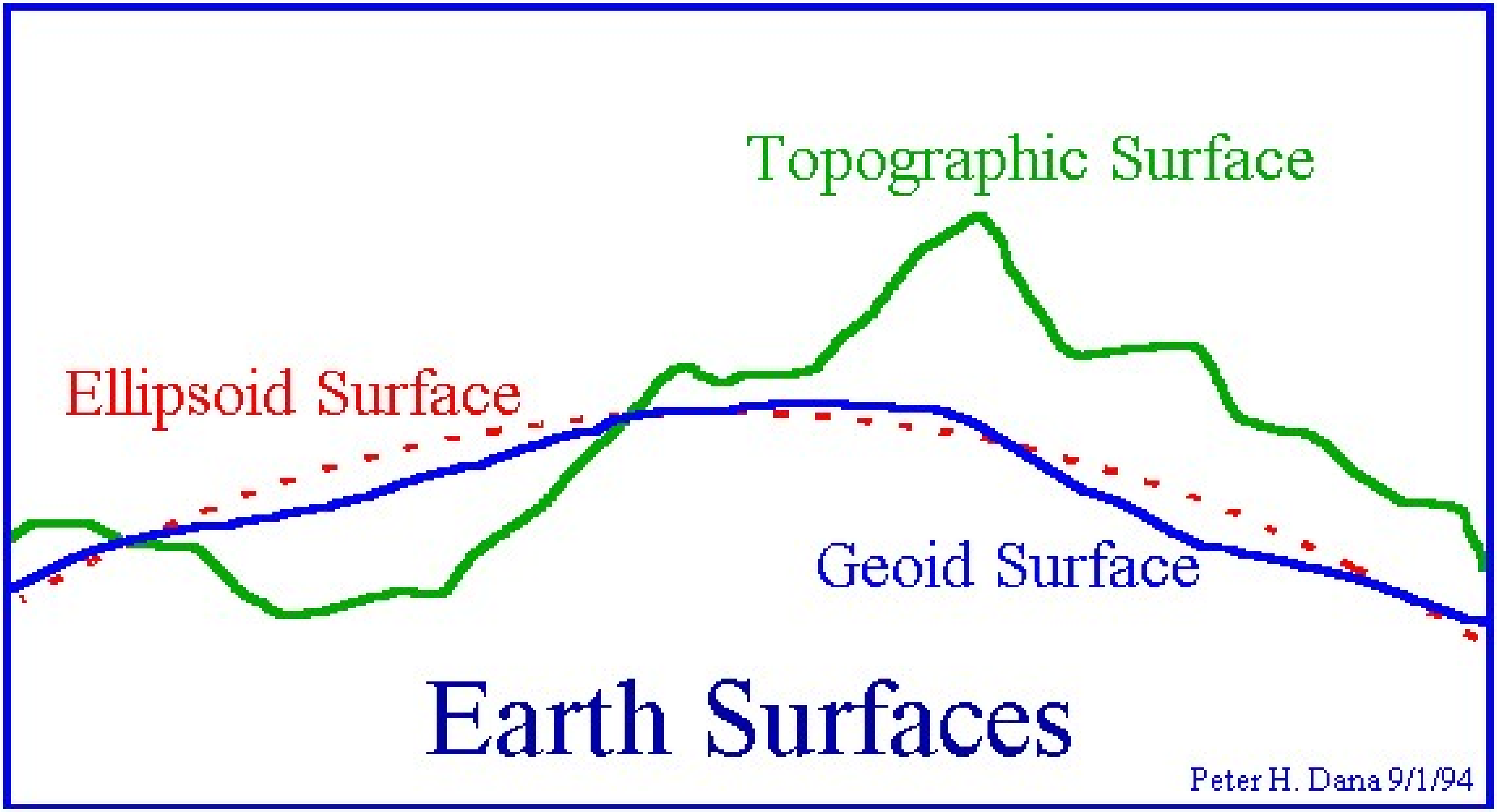
2. Global Geodetic Reference System

- Until the middle of 20th century reference systems were defined and established by each nation using astronomical observations within that nation. So the references varied from nation to nation.
- In late 20th century, the classic geodetic reference systems were replaced by the space geodetic techniques e.g. GPS.
- The centre of Earth, the rotation axis of Earth and the other parameters could be determined with high accuracy by these new technologies. This resulted in the realization of geocentric reference systems which are globally fit.

Global Geodetic Reference System . . . contd.

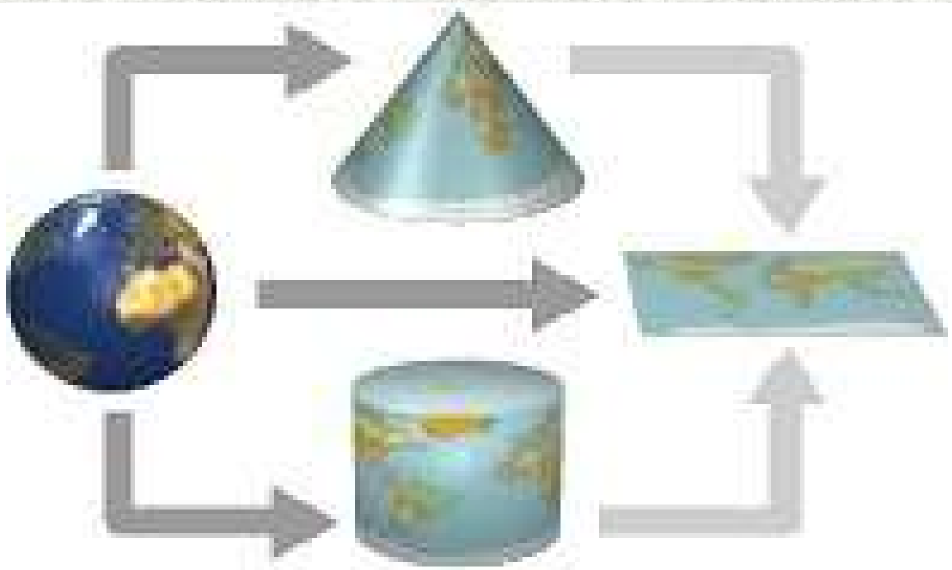
- It has its origin at the centre of mass of Earth
- Z-axis along the assumed rotation axis of Earth
- X-axis along the prime meridian direction
- Y-axis to make a right hand Cartesian coordinate system, which is globally common standard for positioning.

Relationships between the earth's surface, the geoid and a reference ellipsoid



Projection

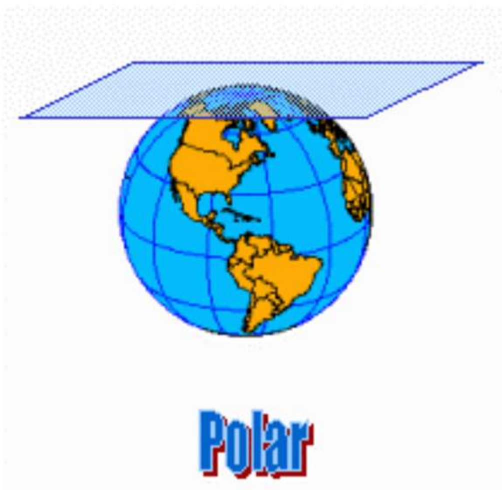
- Transformation of Three Dimensional Space onto a two dimensional map

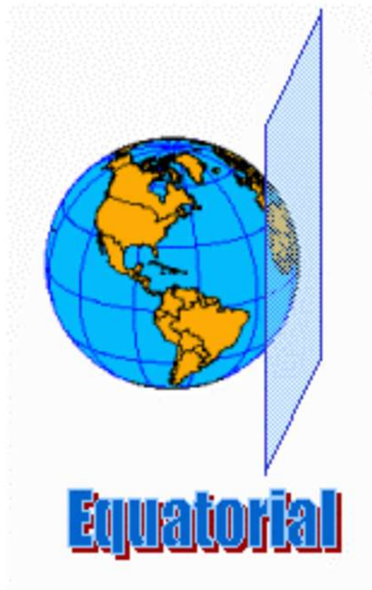


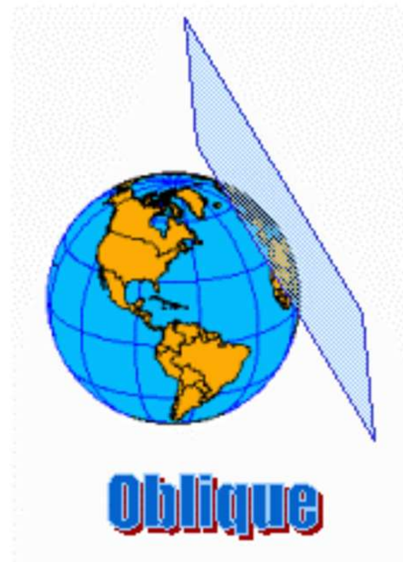
Types of Projections

- Equal Area: maintains accurate relative sizes. Used for maps that show distributions or other phenomena where showing area accurately is important. Examples: Lambert Azimuthal Equal-Area.
- Conformal: maintains angular relationships and accurate shapes over small areas. Used where angular relationships are important, such as for navigational or meteorological charts. Examples: Mercator, Lambert Conformal Conic.
- Equidistant: maintains accurate distances from the center of the projection or along given lines. Used for radio and seismic mapping, and for navigation. Examples: Equidistant Conic, Equirectangular.
- Azimuthal or Zenithal: maintains accurate directions from a given central point. Used for aeronautical charts and other maps where directional relationships are important. Examples: Gnomonic projection, Lambert Azimuthal Equal-Area.

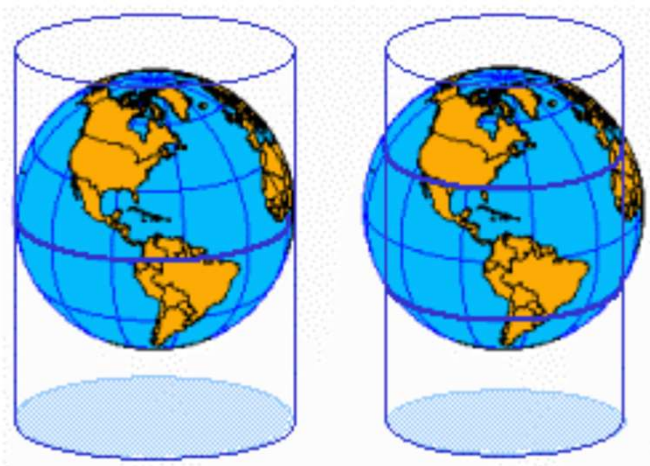
Azimuthal Projections



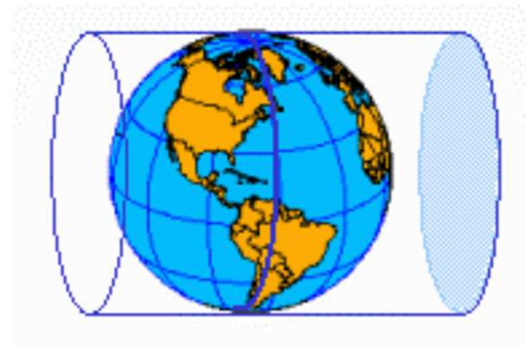




Cylindrical Projections

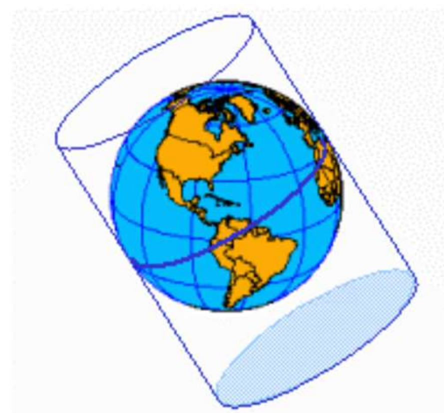


Tangent Secant

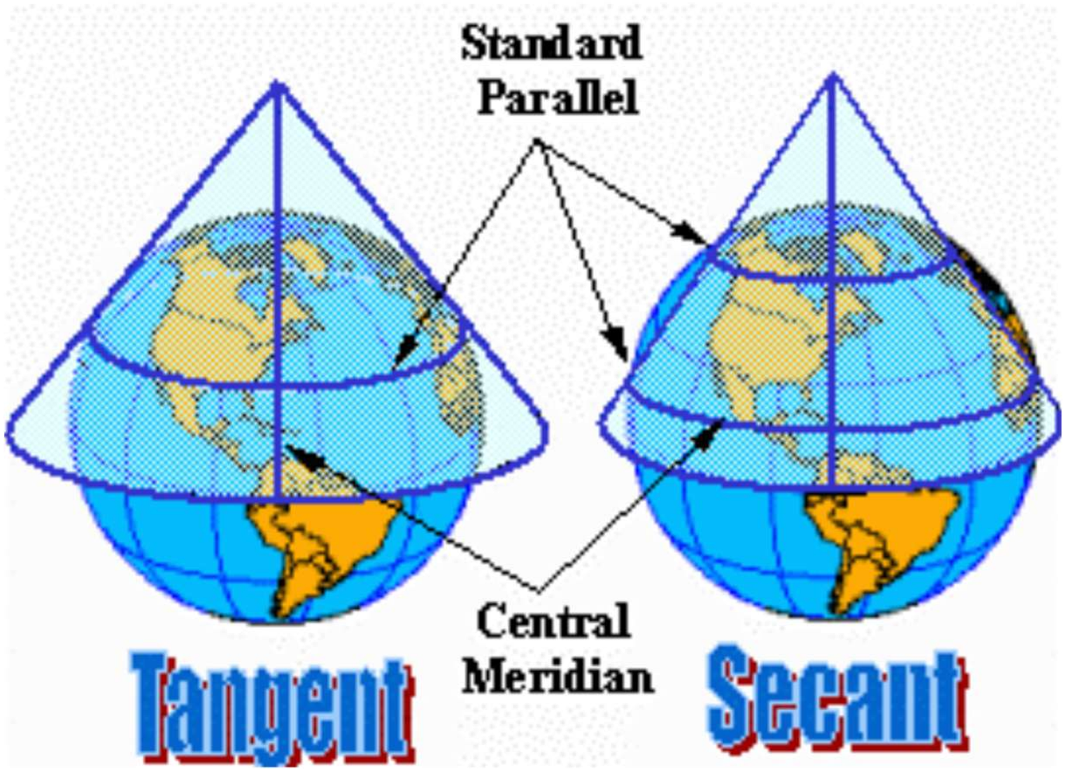


Transverse

Oblique



Conic Projections

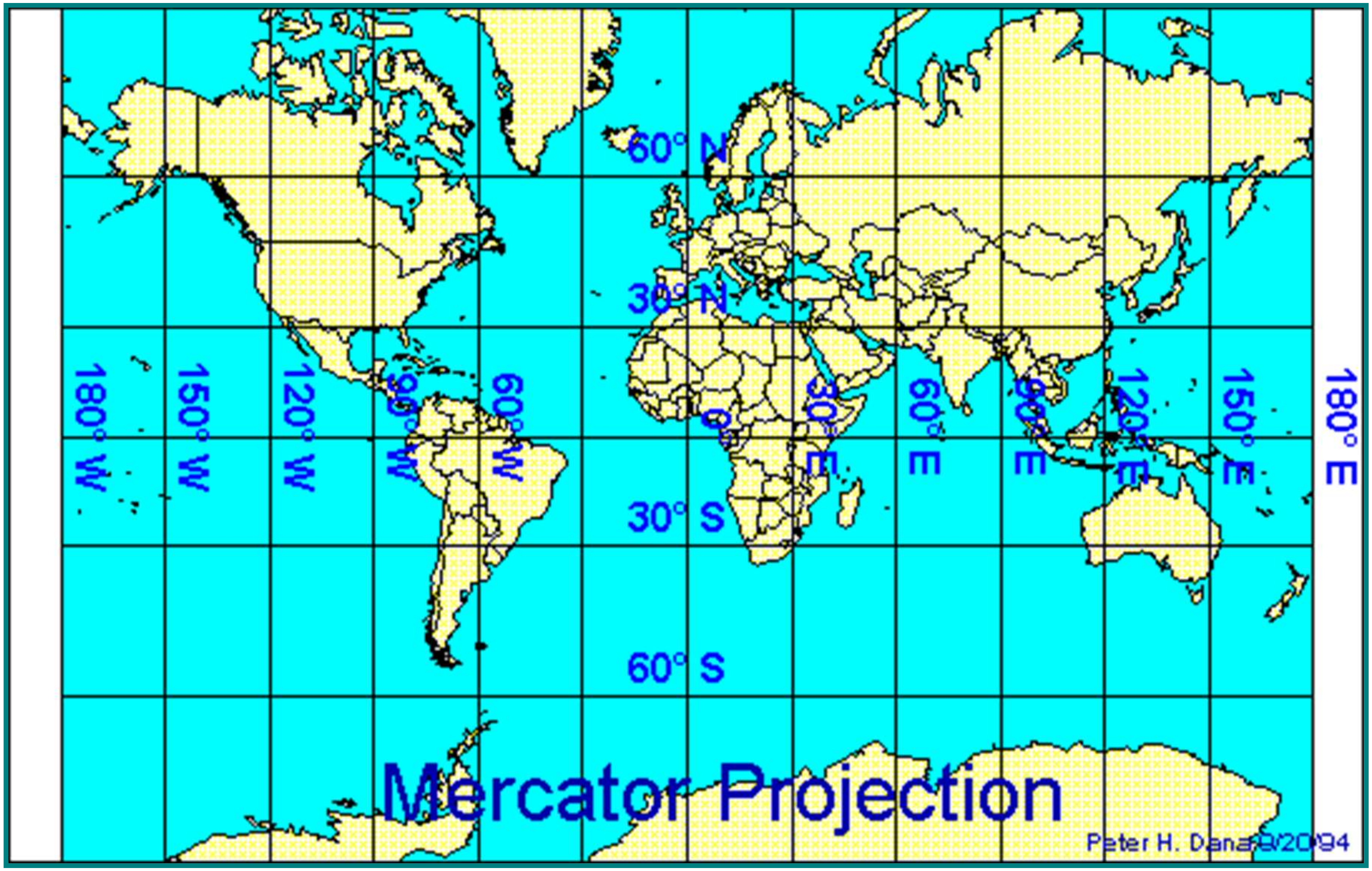


Cylindrical Map Projections

- Cylindrical map projections are made by projecting from the globe onto the surface of an enclosing cylinder, and then unwrapping the cylinder to make a flat surface
  - Mercator
  - Transverse Mercator

Mercator Projection

- Cylindrical, Conformal
- Meridians are equally spaced straight lines
- Parallels are unequally spaced straight lines
- Scale is true along the equator
- Great distortion of area in polar region
- Used for navigation


60° N
30° N
30° S
60° S
180° W
150° W
120° W
90° W
60° W
0°
30° E
60° E
90° E
120° E
150° E
180° E
Mercator Projection
Peter H. Dana 9/20/94

Mercator Projection and the "Greenland Problem"

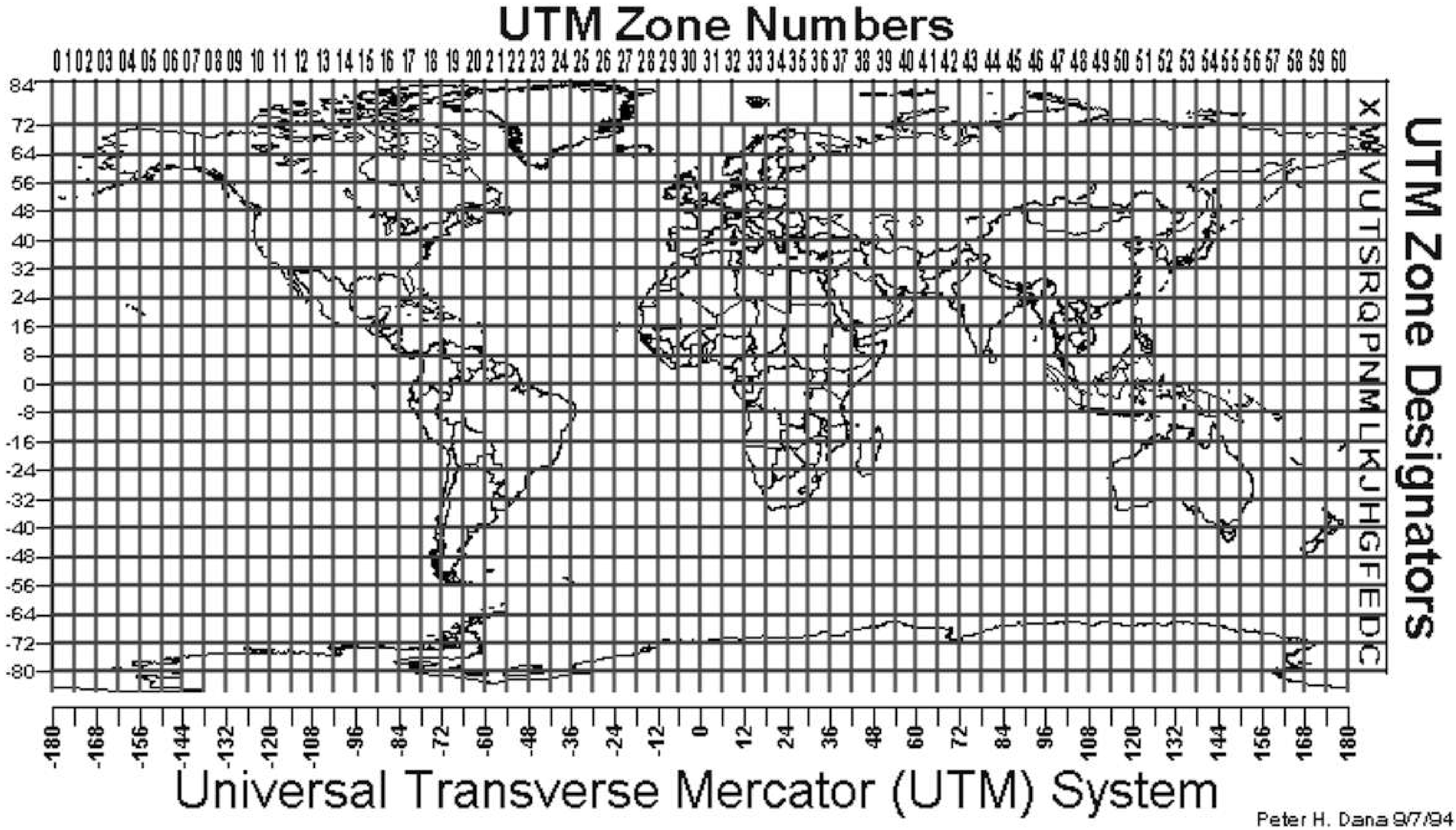
Also known as Northern Hemisphere dominant projection

Transverse Mercator Projection

- Cylindrical (Transverse)
- Conformal
- Central meridian and equator are straight lines
- Other meridians and parallels are complex curves
- Used extensively for quadrangle maps at scales from 1:24,000 to 1:250,000
- For areas with larger north-south extent than east-west extent

Universal Transverse Mercator

- The UTM is a global coordinate system. Particular case of Transverse Mercator Projection.
- The earth between latitudes 84° N and 80° S, is divided into 60, 6° wide;
- Latitude origin – the equator
- Y value UTM northings, increases in northernly direction.
- X value UTM eastings, increases in easterly direction.
- Assumed (false) northing (y): 0 metres for northern hemisphere; 10,000,000 metres for southern hemisphere
- Assumed (false) easting (x): 500,000 metres.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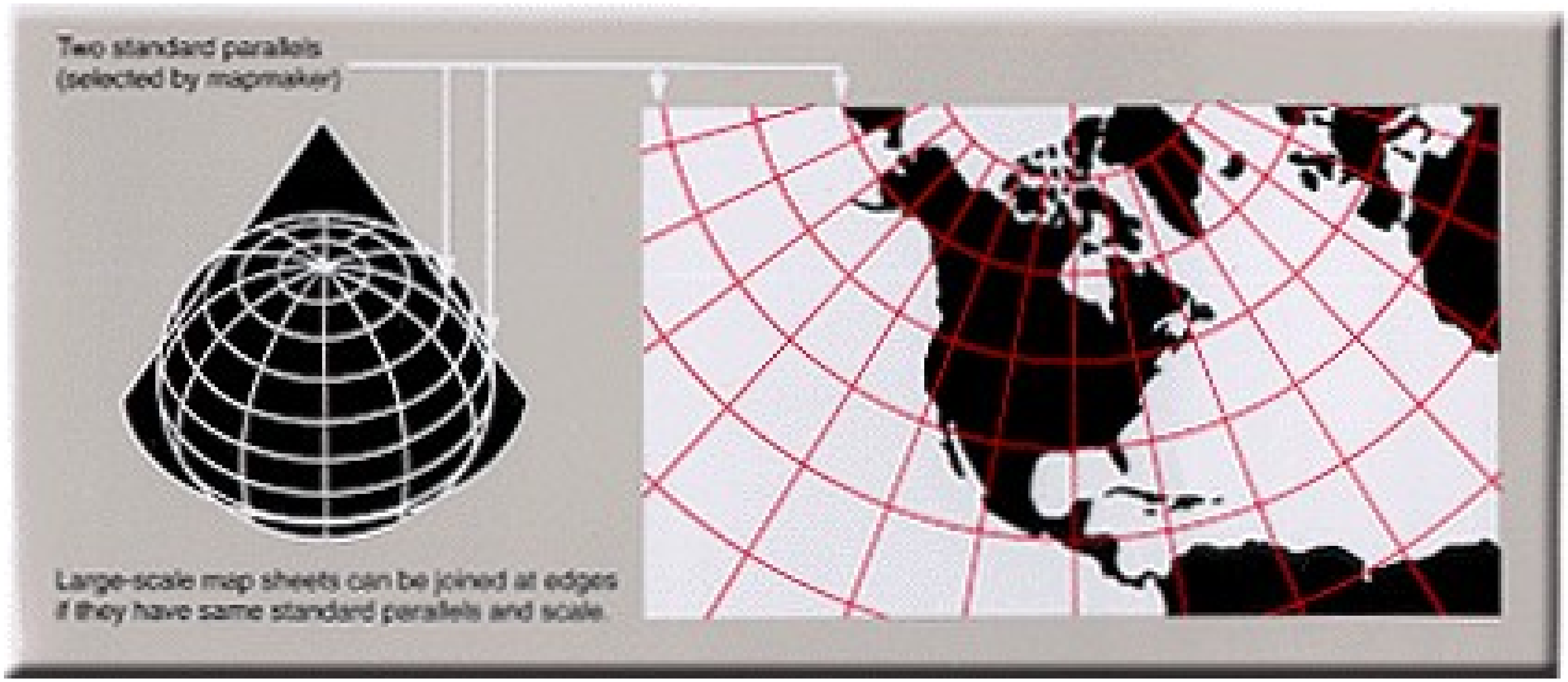
01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
84 72 64 56 48 40 32 24 16 8 0 -8 -16 -24 -32 -40 -48 -56 -64 -72 -80
X W V U T S R Q P N M L K J H G F E D C
UTM Zone Designators
-180 -168 -156 -144 -132 -120 -108 -96 -84 -72 -60 -48 -36 -24 -12 0 12 24 36 48 60 72 84 96 108 120 132 144 156 168 180
Universal Transverse Mercator (UTM) System
Peter H. Dana 9/7/94

Conic Projections

- For a conic projection, the projection surface is cone shaped

- Locations are projected onto the surface of the cone which is then unwrapped and laid flat



LCC PROJECTION

Lambert Conformal Conic Projection

- Conical, Conformal
- Parallels are concentric arcs
- Meridians are straight lines cutting parallels at right angles.
- Scale is true along two standard parallels, normally, or along just one.
- It projects a great circle as a straight line – much better than Mercator
- Used for maps of countries and regions with predominant east west expanse

LCC PROJECTION

Azimuthal Projections

- For an azimuthal, or planar projection, locations are projected forward onto a flat plane.
- The normal aspect for these projections is the North or South Pole.
- Different azimuthal projections, like stereographic, orthographic, and gnomonic, have varying properties and distortions.
- Preserves direction: All azimuthal projections accurately represent true directions (azimuths) radiating outwards from the map's central point. This makes them useful for navigation and applications where knowing direction from a specific point is crucial.
- Symmetrical distortion: Distortions in shape, area, and distance increase as you move away from the central point (point of tangency). The distortions are symmetrical around this center.
- Suitable for specific regions: Azimuthal projections are particularly well-suited for maps of polar regions (when centered at a pole) or areas that are roughly circular in shape and extend equally in all directions from the center.

Choosing a map Projection

- The choice of map projection is made to give the most accurate possible representation of the geographic information, given that some distortion is inevitable. The choice depends on:
- The location
- Shape
- Size of the region to be mapped
- The theme or purpose of the map

NEW MAPPING POLICY

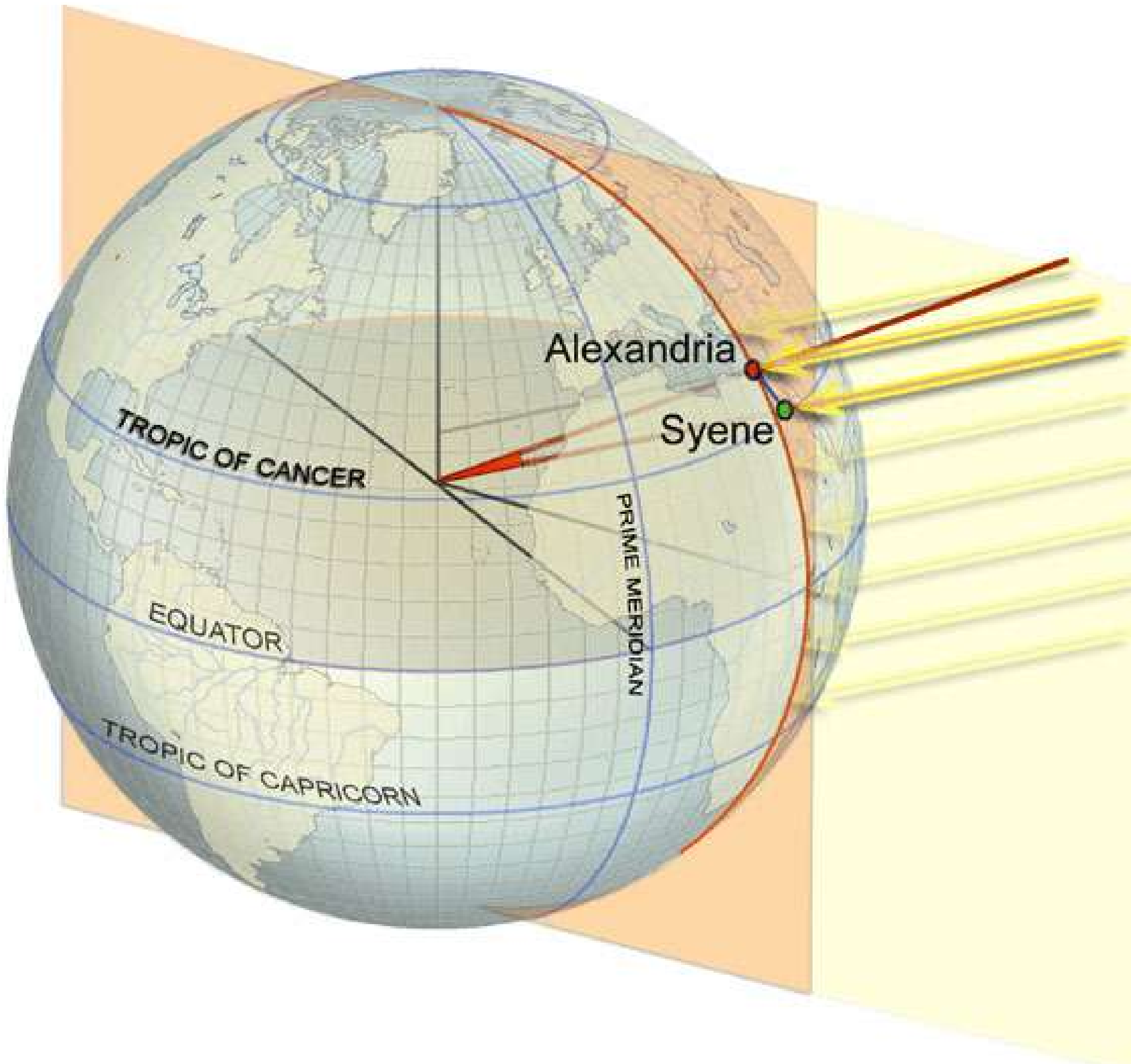
- Open Series Maps
- Defence Series Maps
- Switchover from Everest Co-ordinate System to Geocentric Co-ordinate System (ITRF). ITRF / GRS 80 ellipsoid.
- Survey of India took a conscious decision to go for Geocentric Reference Frame.

Conclusions

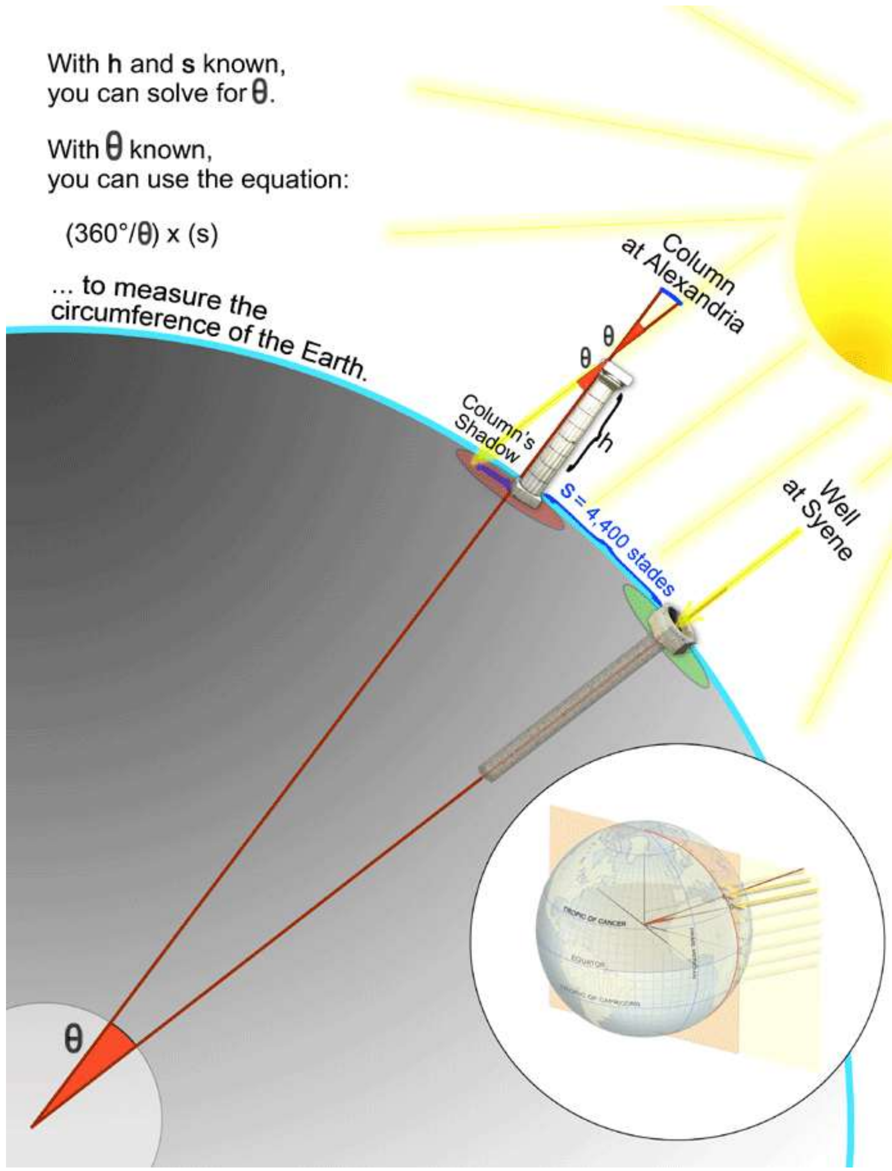
- We need to project geospatial data for any analysis
- Makes it possible to use data from different sources
- Several Projections to choose from
- Projections inevitably distorts at least one property
- Can choose suitable Map Projection
- Can control scale error

References

- **GIS Fundamentals: A First Text on Geographic Information Systems”** by Paul Bolstad
- **Geographic Information Systems and Science”** by Paul A. Longley et al.



Alexandria
Syene
TROPIC OF CANCER
PRIME MERIDIAN
EQUATOR
TROPIC OF CAPRICORN


With h and s known,
you can solve for θ.
With θ known,
you can use the equation:
(360°/θ) x (s)
... to measure the
circumference of the Earth.
Column
at Alexandria
θ
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s = 4,400 stades
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